Stage 5 — main board only. DevKit USB-C programming and main-board USB-C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220-T22D interface: fit one E220-400T22D OR E220-900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull-downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input-only, no pull-down.

OLED: GPIO21/SDA and GPIO22/SCL are signal-only reservations; OLED VCC and I2C pull-ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105-GF-A: all A/B VBUS, GND and shell contacts are explicit; D+/D- are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 -> SN74AHCT1G125DBVR (OE=GND, 5V_SYS) -> WS2812B-V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE. TPD4S311 CC protection is not instantiated: its mandatory 2.7-4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 M0; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test-point footprints/locations are deferred.